

Master of Science
in

Knowledge and Technology Management



THE HONG KONG
POLYTECHNIC UNIVERSITY

香港理工大學

Department of Industrial
& Systems Engineering

Managing Knowledge (ISE 542)

Lecture 1: The Evolution of Knowledge



Learning Outcome

- To learn the evolution of the concept of knowledge
- To understand and differentiate various types of knowledge
- To explain the way in which knowledge can be converted (Nonaka's cycle of knowledge conversion)
- To learn the concept of metaphors and levels of knowledge
- To familiarize with the critical elements of KM



Concept of Knowledge

The Concept of Knowledge

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Timeline: Antiquity 4000 - 2400BC

Definition of knowledge: ?

Knowledge model: Narratives and symbolic depictions

Important discovery: No written record



Antiquity
4000-2400BC



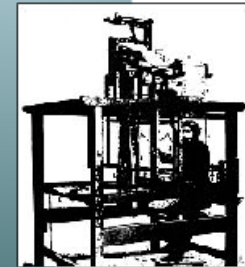
Greek
Philosophers
800BC-500AD



Medieval/
Middle Age
500AD-1400s



Modern Era
1900 onwards



Industrial
revolution
1500s-1800s

Concept of Knowledge

The Concept of Knowledge

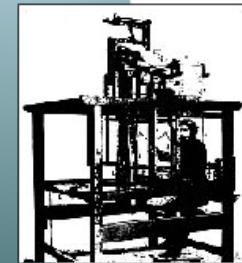
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Timeline: Greek Philosophers 800BC - 500AD

Definition of Knowledge: Knowledge is justified true belief

Knowledge model: Mathematics and logic

Important discovery: Geometry, astronomy, and Aristotelian logical evidence



Concept of Knowledge

The Concept of Knowledge

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Timeline: Medieval / Middle Age 500AD - 1400s

Definition of Knowledge: Knowledge is profess and apprentice

Knowledge model: Scholastic method

Important discovery: Practical manuals and formulation of knowledge that can still be found at present



Antiquity
4000-2400BC



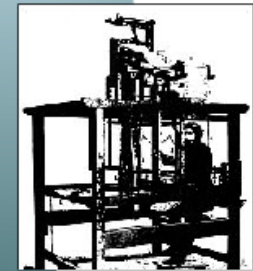
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Modern Era
1900 onwards



Industrial
revolution
1500s-1800s



Concept of Knowledge

The Concept of Knowledge

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Timeline: Industrial revolution 1500s - 1800s

Definition of Knowledge: Knowledge is power empowering effective action

Knowledge model: Scientific method

Important discovery: Testing knowledge by proven statistical means for validation



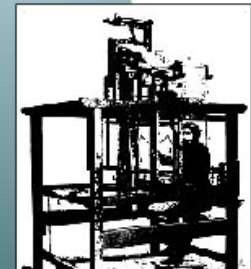
Antiquity
4000-2400BC



Greek
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Medieval/
Middle Age
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Industrial
revolution
1500s-1800s



Modern Era
1900 onwards



Concept of Knowledge

The Concept of Knowledge

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Timeline: Modern Era 1900 onwards

Definition of Knowledge: A part of the market-economy system

Knowledge model: Learning organization

Important discovery: Knowledge arises through interpretation or experience, abstraction, and conceptualisation



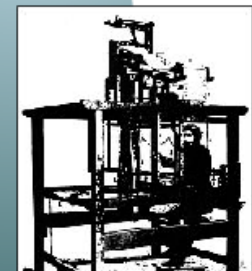
Antiquity
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Greek
Philosophers
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Medieval/
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Industrial
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Modern Era
1900 onwards



Data, Information & Knowledge

Data

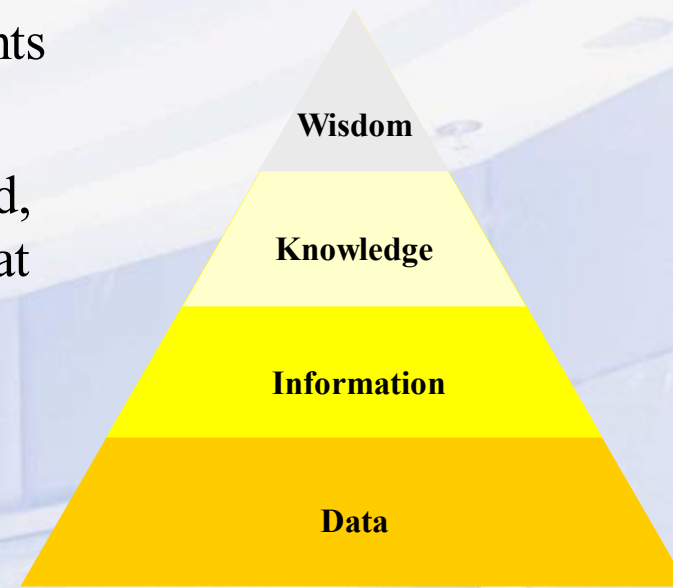
- Data is a set of discrete, objective facts about events

Information

- Processed (contextualised, categorised, calculated, corrected, and condensed) data to show & display what is going on

Knowledge

- Organised/synthesised information to explain or predict events & new situations, to make connections between what is happening now and what happens next, and to make decisions
- Knowledge is often a mix of contextual information, framed experience, expert insight values & beliefs that provides a framework for evaluating & incorporating new experiences & information



The Knowledge Pyramid



In-class Exercise 1: Case Study

The outbreak of the Severe Acute Respiratory Syndrome (SARS) has caused much public concern. The issues have been reported through various channels such as TV, radio, internet, newspapers, etc. Hong Kong has introduced a rigorous contact tracing procedure and good practice. All close contacts of known SARS cases are quarantined at home. In addition, their Hong Kong Identity Card numbers are passed to the Immigration Department to ensure that such people cannot leave the territory. Contact tracing has established epidemiological links for the majority of Hong Kong's SARS patients. Around 8.6% of patients have no identifiable exposure source. Detailed investigation is ongoing.

As of today, a total increase of 168 new cases and 26 deaths have been reported. The distribution of the new deaths were reported in China (17 cases), HKSAR (6 cases), Taiwan, China (2 Cases) and Singapore (1 Case).”

(Source: http://www.who.int/csr/don/2003_05_12/en)

Discussion question

Based on the above scenario, identify the difference between data, information, and knowledge. Give an example for each ONE of them.



Explicit vs. Tacit Knowledge

Explicit knowledge

- Easier to identify
- Re-usable in consistent and repeatable manner - for decision-making and/or exercising judgement
- Can be stored as a written procedure or as a process in a computer system
- Stored as artefacts - artificial physical or virtual entity that can be measured, identified, distributed and audited

Tacit knowledge

- Lessons learned, methodologies, cases, stories
- Personal, context-specific, hard to formalise and communicate
- Intuitions, rules of thumb, mind-sets, unwritten rules of turf and territory, unconscious values
- Trivial fundamental philosophy

In-class Exercise 2

Which of the following is/are the right combinations of examples of an “Explicit Knowledge” and a “Tacit Knowledge”?

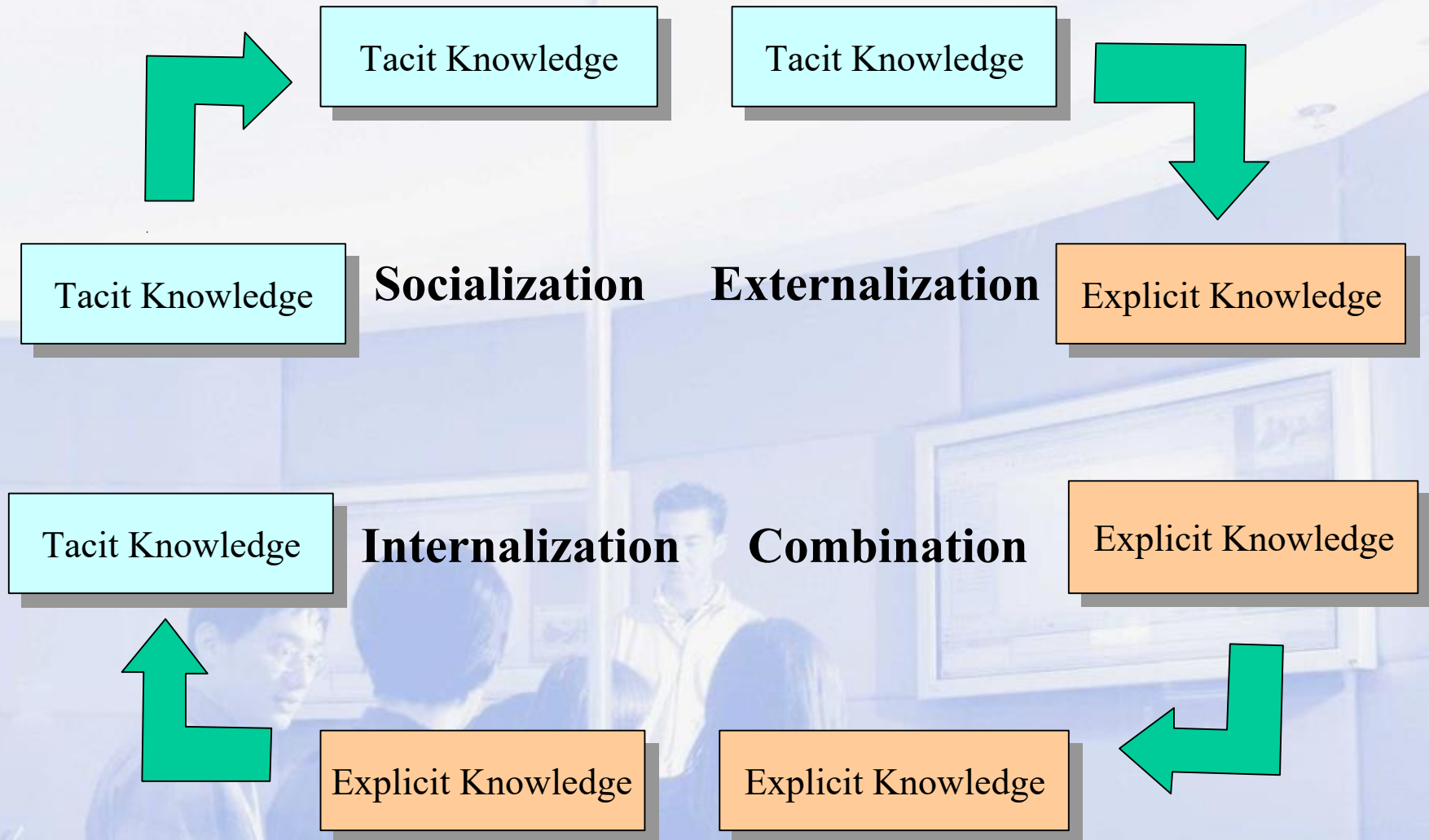
	<u>Explicit Knowledge</u>	<u>Tacit Knowledge</u>
I.	Mentality of customer service staff	Customer databases
II.	Audio records for site visit	Training experience
III.	Story book	Story telling skill
IV.	Massive open online course (MOOC) materials	Video records in MOOC platform

Select the best answer from the following:

- A. I and II
- B. II and III
- C. II and IV
- D. III and I
- E. III and IV



Nonaka's Four Modes of Knowledge Conversion

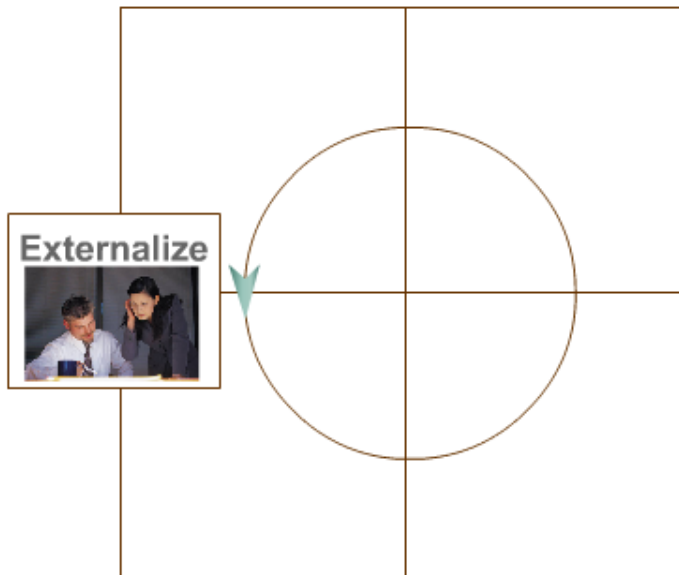


Nonaka's Four Modes of Knowledge Conversion (SECI Cycle)

Nonaka's cycle of knowledge conversion

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Externalization



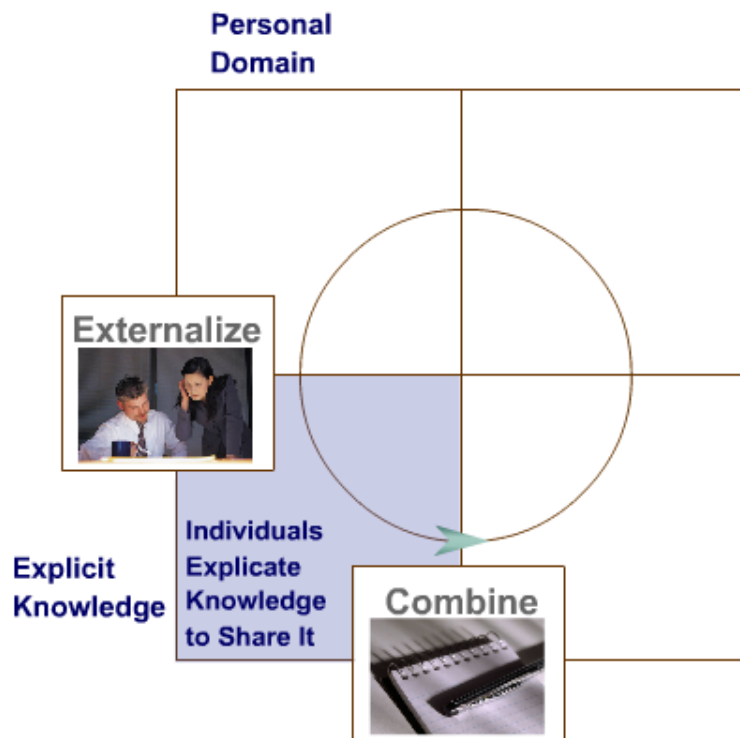
The process for making tacit knowledge explicit is externalization. One case is the articulation of one's own tacit knowledge and the other is translating the tacit knowledge of others into a readily under-standable form. During dialogue, people can share beliefs and learn how to articulate their thinking better, through instantaneous feedback and the simultaneous exchange of ideas. Externalization is a process among individuals within a group. E.g. writing a case study and reports.



Nonaka's Four Modes of Knowledge Conversion (SECI Cycle)

Nonaka's cycle of knowledge conversion

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Combination

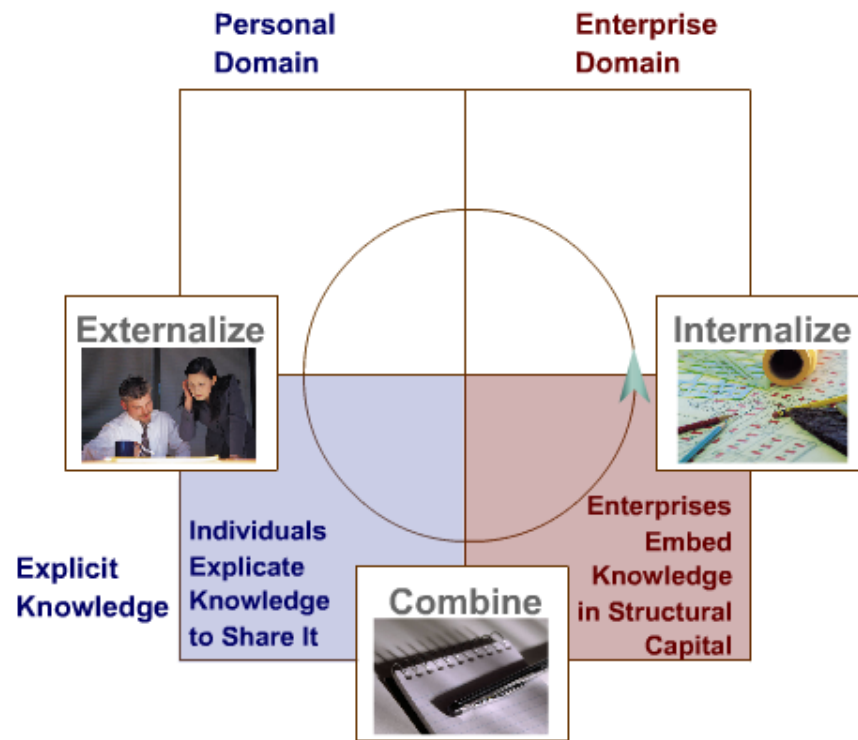
Combination is a process of converting explicit knowledge into a more usable means of explicit knowledge. Since explicit knowledge can be conveyed in documents, emails, databases, it can be made more usable by collecting, disseminating and editing/processing it. This is the area where information technology is most helpful. Combination allows knowledge transfer among groups across organizations. E.g. document management system.



Nonaka's Four Modes of Knowledge Conversion (SECI Cycle)

Nonaka's cycle of knowledge conversion

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Internalization

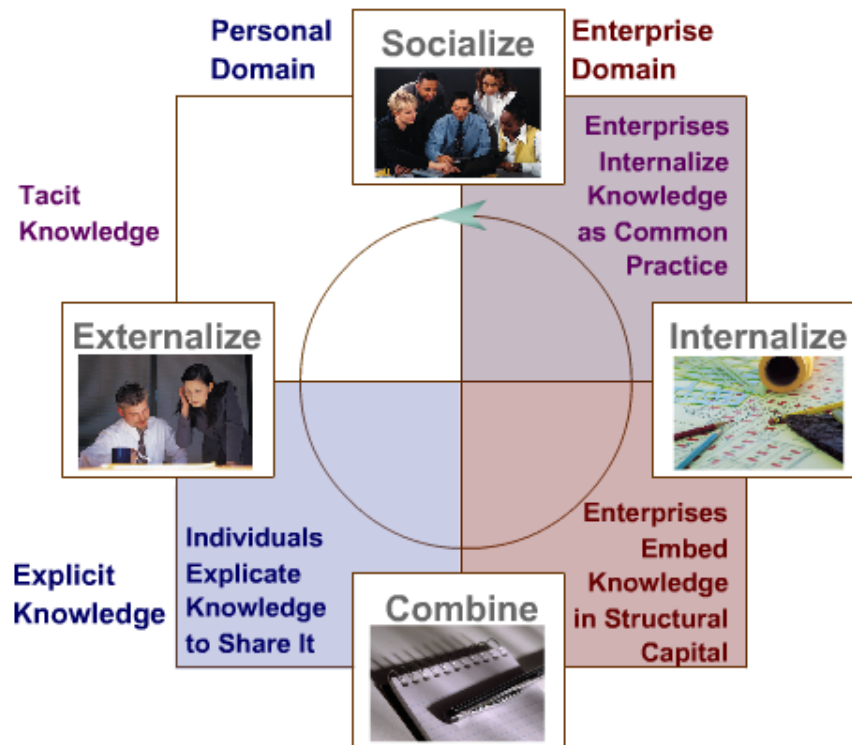
Internalization is a process of understanding and absorbing explicit knowledge thus turning it into tacit knowledge held by the individual. Knowledge in the tacit form is actionable by the owner via actual doing or through simulations. The internalization process transfers organization and group explicit knowledge to the individual. E.g. learning a skill through training and practice.



Nonaka's Four Modes of Knowledge Conversion (SECI Cycle)

Nonaka's cycle of knowledge conversion

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Socialization

The process that transfers tacit knowledge in one person to tacit knowledge in another person is socialization. It is primarily a process between individuals. It involves capturing knowledge by direct interaction and sharing experience with individuals outside and inside an organization. This results in acquired skills and common mental models. E.g. apprenticeship, story telling and experience sharing from lessons learned.



Nonaka's Four Modes of Knowledge Conversion (SECI Cycle)

Nonaka's cycle of knowledge conversion



When viewed as a continuous learning process, the model becomes a clockwise spiral. Organizational learning depends on initiating and sustaining the learning spiral. The model is a spiral but not a cycle. This is because as one "Learns" around the spiral, understanding moves to deeper and deeper levels.



In-class Exercise 3

According to Nonaka's Four Modes of Knowledge Conversion, classify the following items into the categories of externalization, socialization, combination and internalization:

- I. Student reads and studies teaching materials in a massive open on-line course (MOOC) related to knowledge management.
- II. Students discuss project ideas in the tutorial class.
- III. Some software applications tools are stored, indexed and made available for download in an Enterprise Portal.
- IV. Writing a standard operating procedures (SOP) documents for warehouse management.

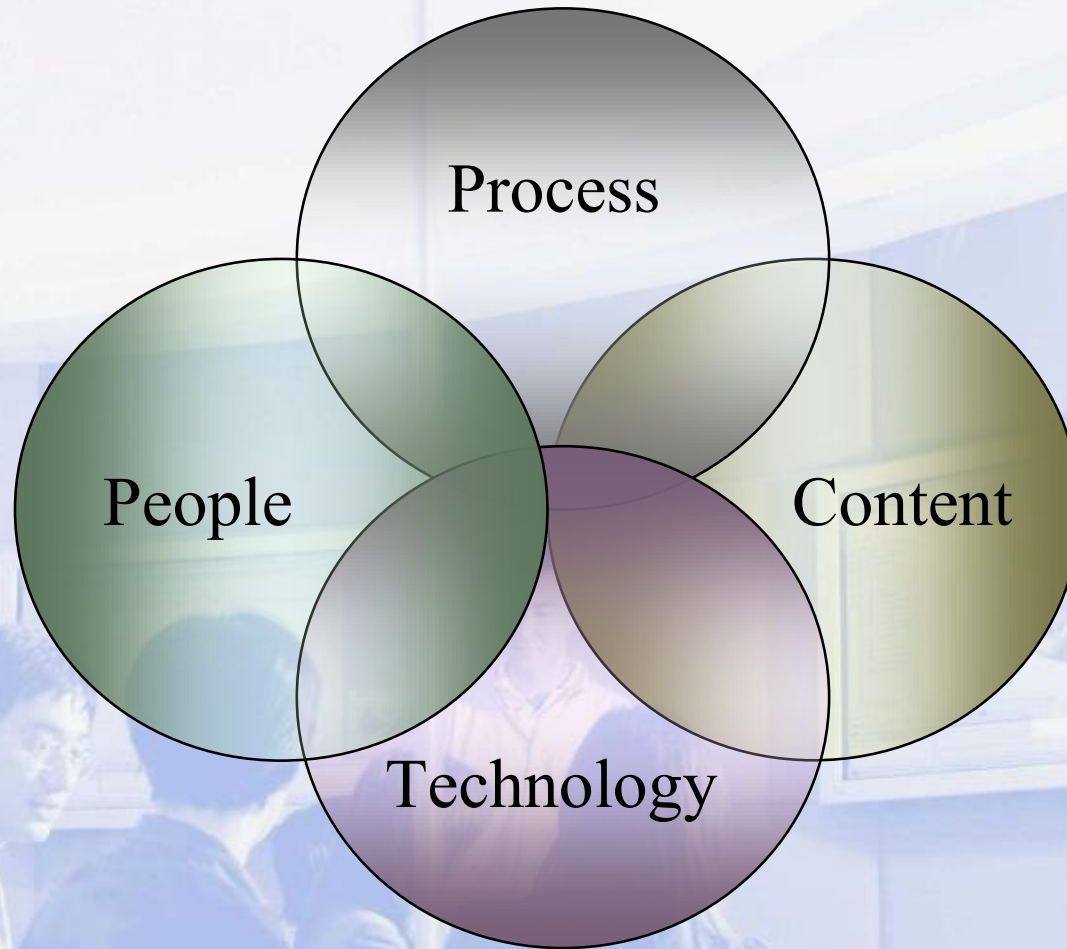
Select the best answer from the following:

	<u>Socialization</u>	<u>Externalization</u>	<u>Combination</u>	<u>Internalization</u>
A.	I	II	III	IV
B.	I	III	IV	II
C.	II	IV	III	I
D.	II	IV	I	III
E.	None of the above			

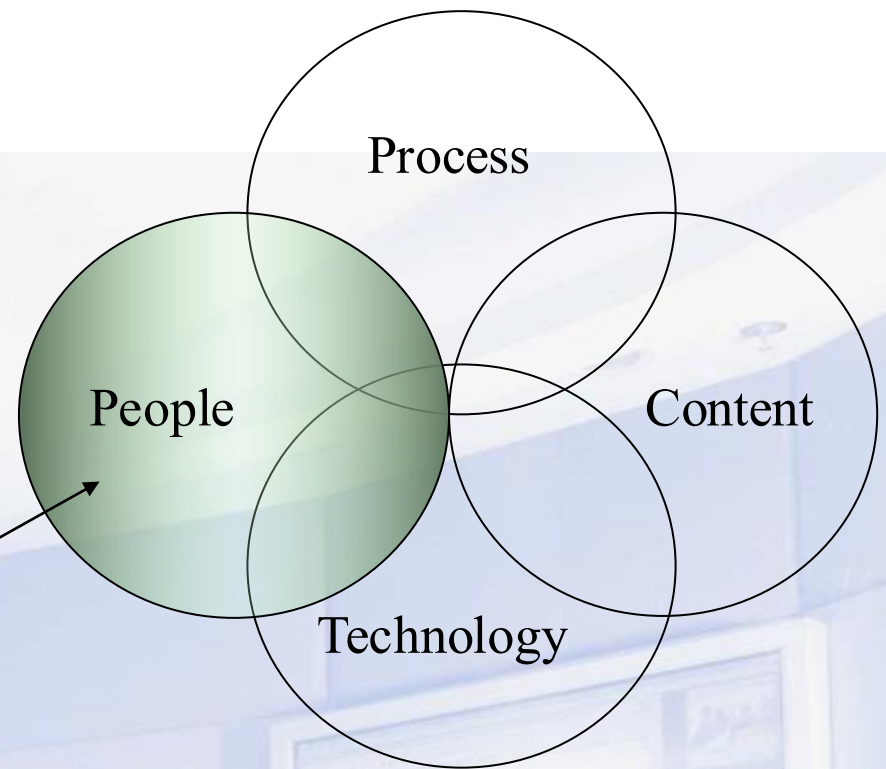


Elements for KM

KM has as its four elements all wrapped together:



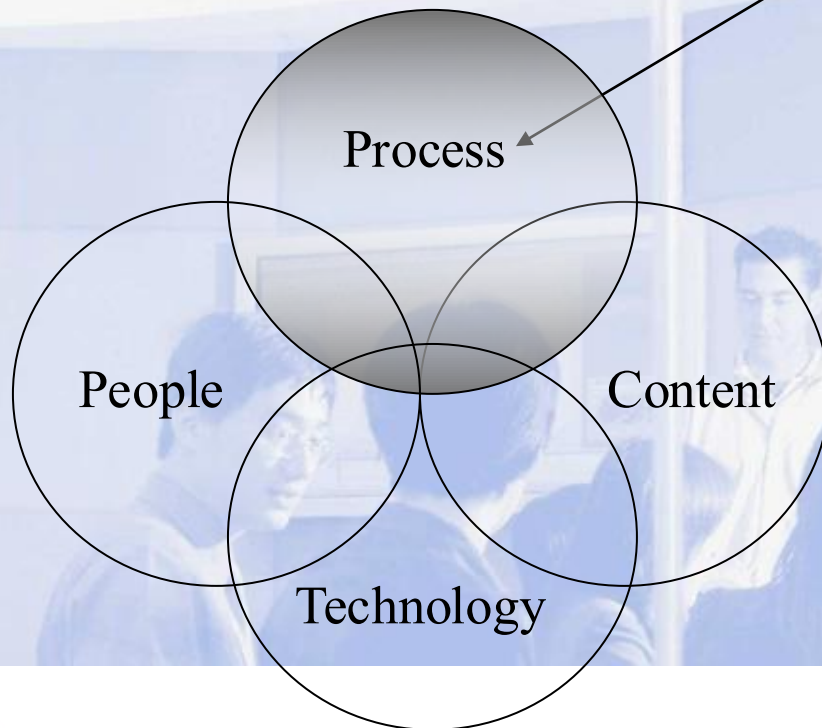
People element for KM



- i. The people element encompasses a diverse range of organizational, cultural and personnel issues which consist of organizational complexity, leadership, learning and organizational design.
- ii. Leadership promotes and values the sharing and use of knowledge. Organizational design facilitates team building and collaboration.
- iii. The people element drives goals such as building a learning environment, knowledge sharing and innovation through collaboration.



- i. The process element encompasses a range of qualitative and quantitative approaches to facilitate knowledge creation, and to capture, validate, transfer, and disseminate knowledge throughout the organisation.
- ii. A good understanding of the way that knowledge can be embedded in business processes and supported by knowledge management is vital for enhancing the organizational performance.

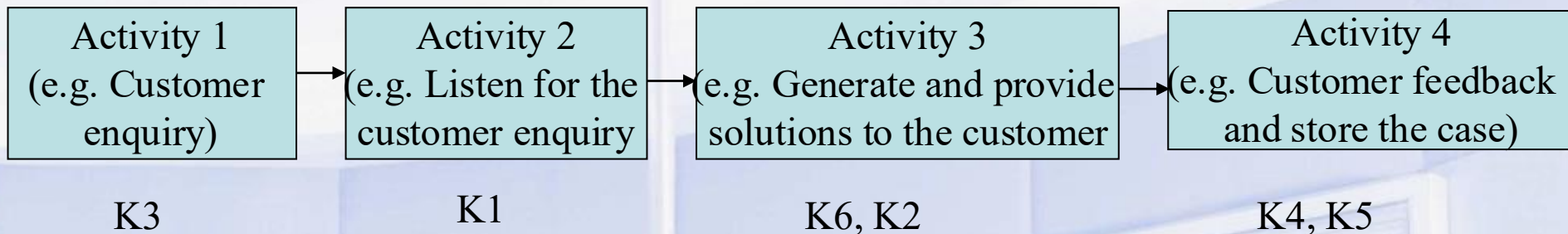


Process element for KM



Business process embedded with Knowledge processes

Various knowledge processes are embedded in different activities in the business process



- K1- Knowledge Process 1 (e.g. Knowledge acquisition)
- K2 – Knowledge Process 2 (e.g. Knowledge transfer)
- K3 – Knowledge Process 3 (e.g. Knowledge sharing)
- K4 – Knowledge Process 4 (e.g. Knowledge retention)
- K5 – Knowledge Process 5 (e.g. Knowledge validation)
- K6 – Knowledge Process 6 (e.g. Knowledge generation)



In-class Exercise 4 – Scenario Case Study

Scenario

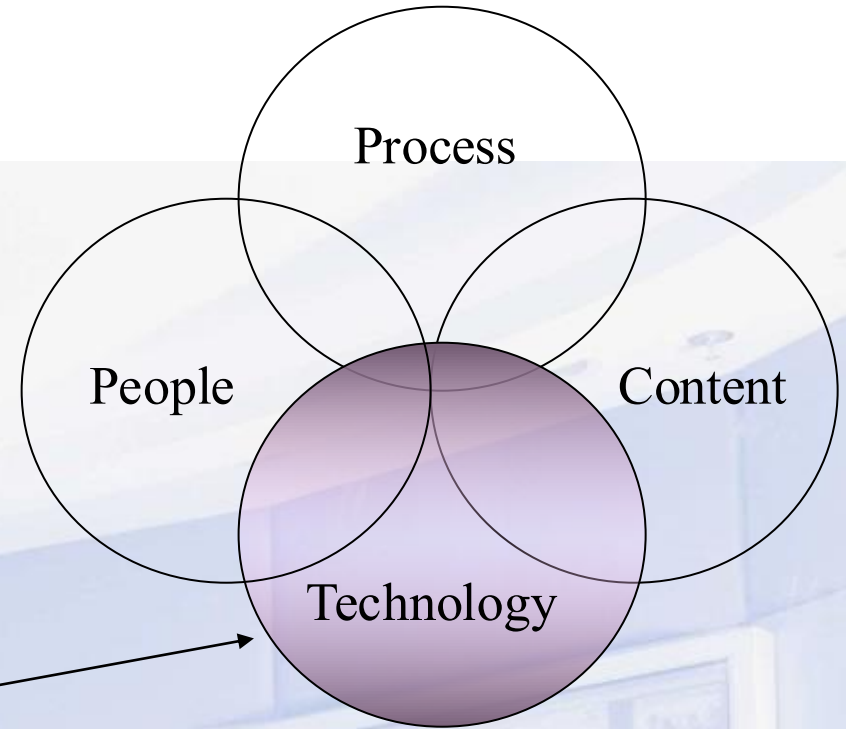
ABC is a toy manufacturer. In promoting their products, ABC created a Facebook page for customers to post reviews of products. ABC possesses its own social media database which captures customer information. All data and records are categorized and indexed in the social media database so that different teams of customer service staff can reuse them to solve the customer problems.

Discussion question

- (i) Discuss the business process in ABC and explain how the knowledge processes are embedded in the business process.



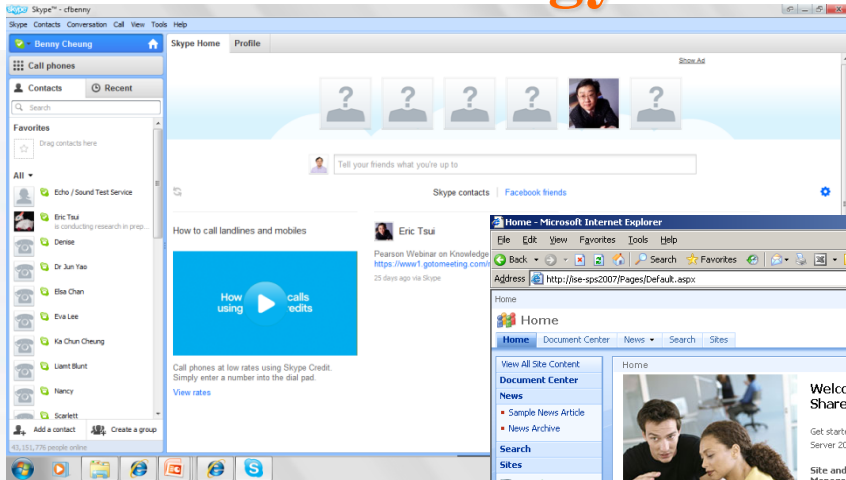
Technology element for KM



- i. Knowledge management is enabled by employing appropriate technologies.
- ii. A standardized and consistent, reliable technological infrastructure and enabling technologies can be used as tools to establish the environment for KM.
- iii. Technologies that enable KM include groupware, databases, knowledge portals, search engines, data mining systems, knowledge-based systems, collaborative tools, etc.



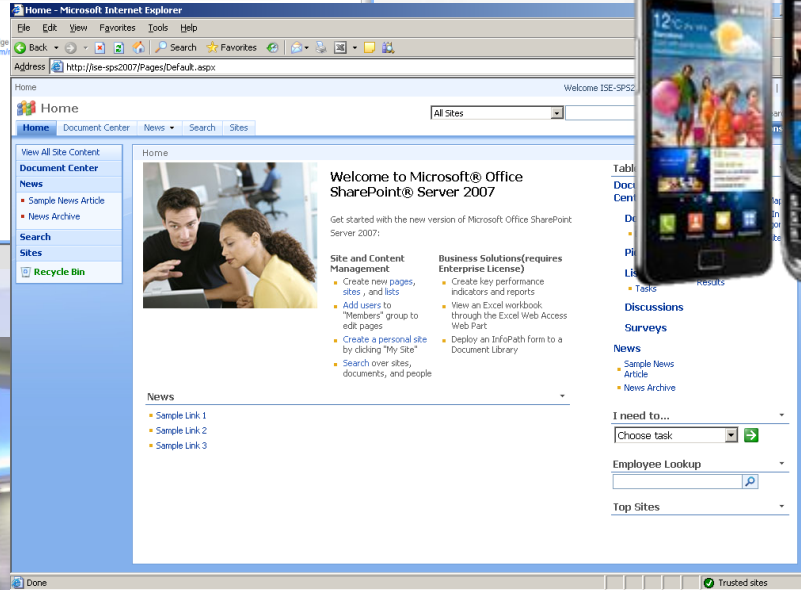
Technology element for KM



Collaborative Tool



Physical KM Environment



Knowledge Portal



Mobile technology

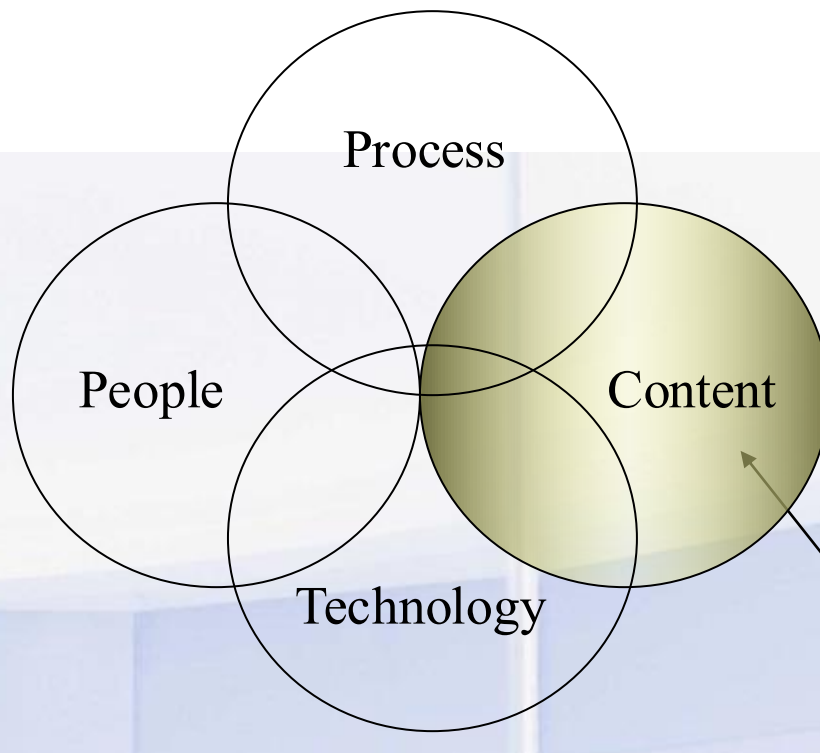


Knowledge Repository



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Systems Engineering

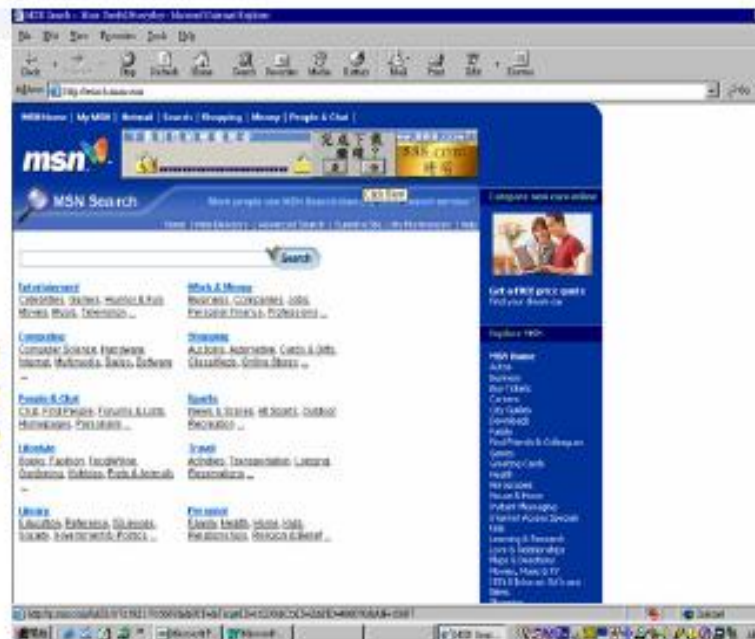


Content element for KM

- i. The content element refers to the characteristics of knowledge, the structure, its formats and organization.
- ii. Content is applicable to both explicit and tacit knowledge content.
- iii. It is related to the development and leveraging of intellectual capital, knowledge sharing and innovation through collaboration.



Examples of content management



Classification of content for portal

Part 3. Process and practice

1. What are the THREE most critical processes within your department or in your job area?

- _____
- _____
- _____

(Example: Order processing can be a process, things need to be carried out like receiving customer queries, entering the transaction data, working the customer on, can be tasks in an accepted process)

2. What are the most difficult task(s) in your job area?

3. Can you think of any specific areas within your job area that would benefit from improving collaboration working with personnel either inside or outside the company?

4. How would you measure the effectiveness of the process stated in the previous question?

Example:	
Accuracy of _____	Timeliness of _____
Cycle time of _____	Cost saving of _____
Decision quality of _____	Rapid response to _____

5. How often do you rely on past documents for future decision making or doing new projects?

Never Rarely Occasionally Often Always

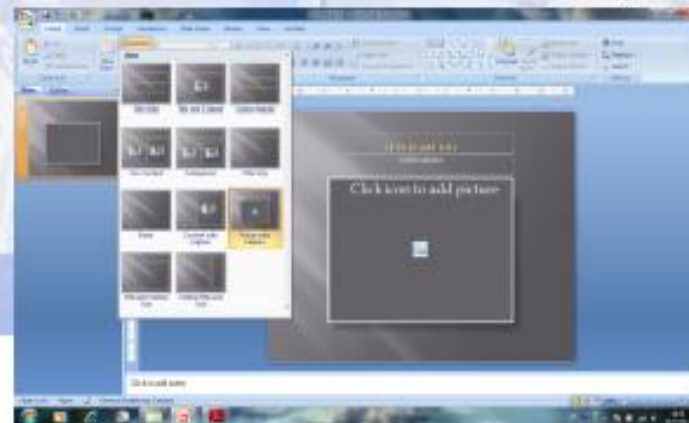
6. Do you keep records on what you have done?

Yes No

If Yes, In what format e.g. document, email, etc. _____
Where they are located _____

2

Designing Template



Organize content for presentation

Organization for content for shared folders

In-class Exercise

Classify the following items into people, process, technology, and content of KM. Briefly explain the answers.

- I. On-job training course is provided for new staff to process customer transaction records in sales and marketing.
- II. Telephone manner and experience of the customer service staff.
- III. Layout of the sales order form.
- IV. Document management system.



Metaphors for KM

- A creatively designed metaphor can depict a million words and provide a good insight into a complex process.
- Metaphors can be designed to explain the abstract concept of KM. It helps to **reduce the level of abstraction** for abstract concepts.
- Metaphor rationalizes the much needed and comprehensive KM process or methodology for increased likelihood of KM success.



Three steps for guiding the construction of metaphor

- **Step 1:** Identify the original concept which aims to reduce the level of abstraction
- **Step 2:** Identify relevant concept which has lower level of abstraction
- **Step 3:** Associate the concept identified in Step 1 and Step 2



Example 1: Water Metaphor

- **Original Concept:**
- Different nature of knowledge

Relevant Concept:

Explicit knowledge

Liquid water

Tacit Knowledge

Water vapor



Example 2: Iceberg Metaphor

Original Concept

Relevant Concept

Explicit Knowledge

Tacit Knowledge

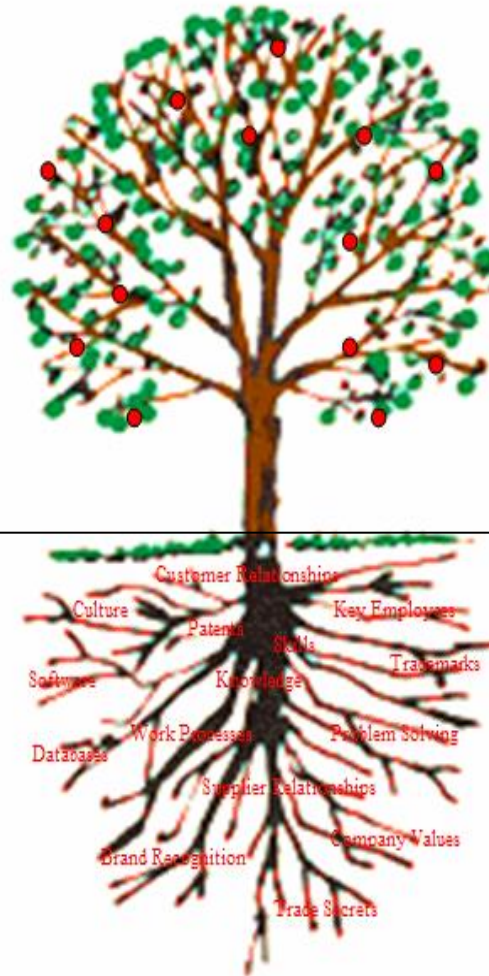


Example 3: Tree Growth Metaphor

Original concept: Power of KM for generating business performance

Relevant concept: Nature of tree growth

*For the tree to flourish
and bear high quality fruit
well into the future...*



Performance of an
Organization with good
prospective

KM includes the management of people, process, technology and culture so as to nurture the healthy growth of the organization but they are invisible.

*...it must be nourished
by a strong, healthy root
system*



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In-class Exercise 6 – Formulating Metaphor

(i) Formulate your metaphor to show the power of knowledge

(ii) Write down and present your metaphor

(15 minutes)



Meta-level of Knowledge

- **Knowledge can be viewed in levels.**
- **Meta-level of knowledge**
- **Level 0** is the world (already exists which can not created)
- **Level 1** is knowledge that ones use to interact with the world
- **Level 2** is a reflection on how people produce, acquire and transmit knowledge i.e. KM processes.
- **Level 3** is a reflection on how people are managing knowledge processes i.e. KM.



Example 1: Meta-level of Knowledge for Manufacturing

Meta-level of knowledge - Manufacturing process

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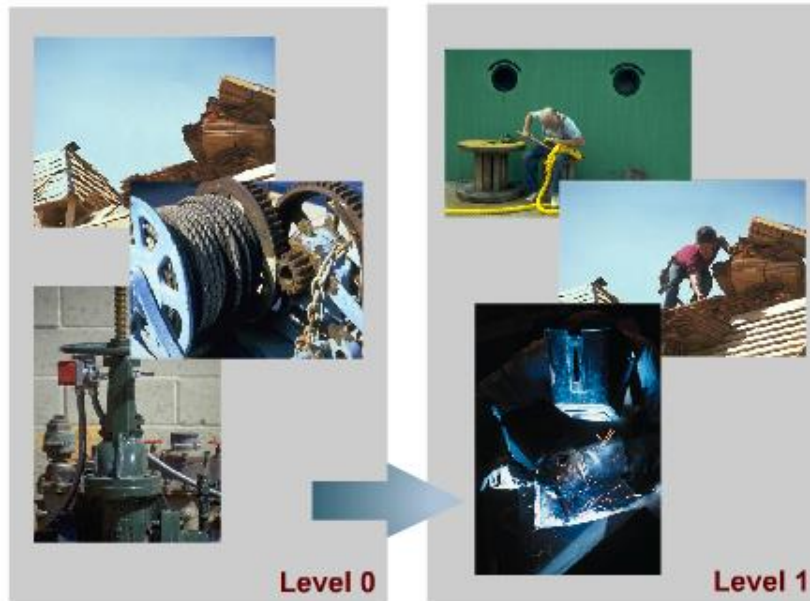
Knowledge at level 0 is represented by the products produced and sold to customers. This includes the workers, machines, parts, raw materials, etc.



Example 1: Meta-level of Knowledge for Manufacturing

Meta-level of knowledge - Manufacturing process

Please Click the  Button



Knowledge at level 1 is where workers are applying their knowledge to manufacture those products and services. The manufacturing process is the firm's knowledge for manufacturing the products.

Example 1: Meta-level of Knowledge for Manufacturing

Meta-level of knowledge - Manufacturing process

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Staff move into knowledge at level 2 when workers or managers reflect on how to improve that manufacturing process.

Example 1: Meta-level of Knowledge for Manufacturing

Meta-level of knowledge - Manufacturing process



Should they reflect on how to improve the process of improving processes, they move into knowledge at level 3 and so on.

In-class Exercise 7

Discuss the meta level of knowledge for customer service management. Briefly explain how different levels of knowledge are articulated to each other.

